

CS381 Homework 2 Problem 1

Connor Couetil

February 16, 2026

1 Exercise 6.5.1

is-half-connected(G): Given a directed graph $G = (V, E)$, returns True if the graph is half-connected, False otherwise.

1. Compute SCCs of G
2. Make new graph, H , where each vertex represents a SCC and edges between those vertices exist if the original graph G has at least one edge between those SCC.
3. Topologically sort H into $SCC_1, SCC_2, \dots, SCC_k$
4. Check if there is an edge between all consecutive SCC .
(SCC_i, SCC_{i+1}) $\in E(H)$ for all $i = 1, \dots, k - 1$
5. Return True if all such edges exist
6. Otherwise return False

Proof: suppose the algorithm returns true. Then each SCC is linked to the next in the order. For vertexes in different SCC, the vertex in the earlier sorted SCC can reach the later SCC by following the chain of edges. If vertexes are in the same SCC, they can reach each other by definition of SCC. Now by contradiction, suppose a graph G is half-connected and the algorithm returns False, implying at least one pair of sorted consecutive SCC does not have an edge between them. Choosing one vertex from each of those SCC would produce a pair of vertices that cannot reach each other, contradicting the half-connected property so if a graph is half-connected, the algorithm will not return false.

Runtime:

1. $O(|V| + |E|)$ to compute SCC of G
2. $O(|E|)$ to build the new graph H
3. $O(|V| + |E|)$ to perform topological sort
4. $O(|V|)$ to check consecutive SCC pairs

Total: $O(|V| + |E|)$

2 Exercise 6.5.2

3 Exercise 5.13

subs(k, h, v): Given parameters k , a set of bids (a bid is a tuple of x , the order price, and y , the inches of sub that fulfills the order), h , the inches of ham sub left, and v , the inches of veggie sub left, returns the maximum profit from selling the two sandwiches to the set of bidders.

1. If k is empty, Return 0
2. $b = k[0]$
3. if b is a bid for ham and $h \geq b_y$
 - A. Return $Max(\begin{array}{l} b_x - \text{cost}(b_y) + \text{subs}(k \setminus b, h - b_y, v), // \text{ Take the bid} \\ \text{subs}(k \setminus b, h, v) // \text{ Don't take the bid} \end{array})$
4. if b is a bid for veggie and $v \geq b_y$
 - A. Return $Max(\begin{array}{l} b_x - \text{cost}(b_y) + \text{subs}(k \setminus b, h, v - b_y), // \text{ Take the bid} \\ \text{subs}(k \setminus b, h, v) // \text{ Don't take the bid} \end{array})$
5. If b is disjunctive and $h \geq b_y$ and $v \geq b_y$
 - A. Return $Max(\begin{array}{l} b_x - \text{cost}(b_y) + \text{subs}(k \setminus b, h - b_y, v), // \text{ Take the ham bid} \\ b_x - \text{cost}(b_y) + \text{subs}(k \setminus b, h, v - b_y), // \text{ Take the veggie bid} \\ \text{subs}(k \setminus b, h, v) // \text{ Don't take the bid} \end{array})$
6. If b is disjunctive and $h \geq b_y$
 - A. Return $Max(\begin{array}{l} b_x - \text{cost}(b_y) + \text{subs}(k \setminus b, h - b_y, v), // \text{ Take the ham bid} \\ \text{subs}(k \setminus b, h, v) // \text{ Don't take the bid} \end{array})$
7. If b is disjunctive and $v \geq b_y$
 - A. Return $Max(\begin{array}{l} b_x - \text{cost}(b_y) + \text{subs}(k \setminus b, h, v - b_y), // \text{ Take the veggie bid} \\ \text{subs}(k \setminus b, h, v) // \text{ Don't take the bid} \end{array})$
8. Return $\text{subs}(k \setminus b, h, v) // \text{ Unable to take bid}$

Given a ham sub m inches long and a veggie sub n inches long, find the maximum profit from selling the subs to a set of k bidders by calling $\text{subs}(k, m, n)$. The total number of calls that will be made is the size of the powerset of the set of bids, so the runtime is $O(2^k)$.